

Hiver SDE Intern Take-Home — Assignment Report

Brand: hulu_support

System: Grounded support agent (intent → retrieve → escalate policy → draft)

Eval set: Golden N=199 (human-labelled) + human reply pack N=50

1. Problem framing — what “good” means

For Hulu Twitter support, a trustworthy agent must:

1. **Route** messages into a small, data-derived intent set (10 intents).
2. **Ground** drafts in historical Hulu replies — not invent prices, refunds, or account actions.
3. **Abstain** (escalate) when evidence is weak, conflicting, account/billing/security-sensitive, or ambiguous — with an explicit reason.

Good ≠ maximum automation. Primary safety metric is **unsafe auto-handle rate** (should-escalate cases that were auto-handled). Coverage is secondary.

What we chose not to build

- End-to-end account tools (password reset, refunds, plan changes)
- Fine-tuned generative LLM as the source of truth
- Mega-brand scope (Amazon/Apple) that explodes the intent space
- Banking77 as primary data (optional only; Hulu TWCS is the product surface)

2. System (short)

```
message → TF-IDF+LogReg intent
          → semantic retrieve (train-only; tfidf_svd embeddings; diversity)
          → evidence strength score
          → deterministic escalation policy (final authority)
          → if auto-handle: copy top historical reply (or LLM draft if keyed)
            else: escalate template + reason
```

Artifacts: data/processed/agent_artifacts/. Demo: `python -m src.cli.run_agent --mode deterministic --message "..."`.

3. Results vs baselines (golden N=199)

| System | Intent acc | Macro F1 | Unsafe auto-handle ↓ | Auto-handle coverage |

| ****Trivial:**** majority intent | 0.126 | 0.022 | 0.000 | 0.000 |

| ****Simple:**** TF-IDF + LogReg | 0.693 | 0.668 | 0.253 | 0.317 |

| TF-IDF retrieve-and-copy | 0.352* | 0.362* | 0.149 | 0.176 |

| ****Ours:**** semantic retrieve + policy (no LLM) | 0.693 | 0.668 | ****0.069**** | 0.166 |

| Semantic + LLM draft | NOT YET MEASURED (no cloud API key in this run) | | |

*Neighbor-intent accuracy for retrieval systems (diagnostic), not a trained classifier.

Human reply quality (N=50, annotator_1) — semantic vs TF-IDF copy:

| | Correctness | Groundedness | Helpfulness |

| TF-IDF retrieve-and-copy | 3.14 | 3.18 | 3.06 |

| Semantic + policy | 2.92 | ****4.86**** | 2.58 |

Semantic is safer/more grounded but over-escalates (~90% of the 50-pack), cutting helpfulness.

LLM-as-judge vs human (N=50, model `ollama:qwen2.5:3b`; no gold labels in judge prompt):

| Dimension | Exact agree | Within ± 1 | Weighted κ | Judge-human bias |

| correctness | 0.36 | 0.72 | 0.15 | -0.66 |

| groundedness | 0.06 | 0.12 | -0.03 | -2.86 |

| helpfulness | 0.32 | 0.82 | 0.21 | +0.84 |

| actionability | 0.38 | 0.88 | 0.10 | +0.14 |

| brand_style | 0.06 | 0.56 | 0.01 | +1.38 |

Read: helpfulness/actionability are usable within-1; groundedness agreement is poor — the local judge under-scores escalate templates that humans rated highly grounded. Do not treat judge scores as a substitute for human labels on groundedness. Full dump: `reports/phase2/judge_human_agreement.json`.

4. Failure analysis — top 5 modes

1. Over-escalation on solvable playback/Live TV issues (n≈91 disagreements)

Example: `gold_2911030` — multi-device buffering; gold says auto-handle; policy escalates.

Hypothesis: Safety-first thresholds + billing/account keyword nets catch too many technical cases; historical copy would have been adequate.

2. Intent confusion between playback vs live_tv vs ambiguous (n≈61)

Example: `gold_1372321` — “Hulu TV live constantly buffering” gold=`playback_error`, pred=`live_tv_issues`.

Hypothesis: Overlapping vocabulary; taxonomy boundaries are real but fuzzy in short tweets.

3. Rare intents underperform (``how_to_feature``, ``service_outage``)

Example: `gold_45990` — concurrent streams on basic package; gold=`how_to_feature`, pred=`billing_subscription` → billing escalate.

Hypothesis: Low train support + lexical overlap with plan/pricing language.

4. Ambiguous / under-specified messages

Example: short complaints without device/error context → `other_ambiguous` → escalate.

Hypothesis: Correct safety behavior, but inflates escalation rate vs gold labels that sometimes allow auto-handle.

5. Off-topic retrieve-and-copy when not escalating

Example: baseline on `gold_284187` (cannot login) retrieved a content/ghost subs reply.

Hypothesis: Lexical retrieval without strong intent gating; diversity helps but does not fix wrong neighbors.

Export: `reports/phase2/failure_analysis.json`.

5. What is misleading about my headline number?

Headline often quoted: TF-IDF+LR / semantic intent accuracy ≈ 0.69 on golden.

Why that number misleads:

1. **Golden labels are single-annotator** (with rule aid). Second annotator agreement is only ~62% intent / ~72% escalate — the ceiling is not 100%.
2. **Accuracy hides rare-class pain** (how_to_feature, service_outage); macro-F1 (~0.67) is the fairer summary.
3. **Intent accuracy $\neq$ reply quality**. Semantic intent matches TF-IDF+LR because it *uses* that classifier; the product risk is reply + escalation.
4. **Unsafe auto-handle 0.069 looks strong** partly because the policy **over-escalates** (coverage only ~17%). Safety improves by refusing work, not by answering better.
5. **Human helpfulness (2.58)** on the 50-pack is *worse* than retrieve-and-copy (3.06) for the same reason: escalate templates are grounded but not helpful.
6. **Historical tweets $\neq$ current Hulu policy**. Groundedness to 2017-era replies can still be operationally wrong today.
7. **No cloud LLM draft eval** in the headline path — do not treat deterministic copy as “LLM agent quality.”

6. What I'd do with one more week

1. Calibrate intent confidence; retune escalation on validation for fewer over-escalations without raising unsafe auto-handle.
2. Intent-specific policies (playback auto-handle freer; billing/account always escalate).
3. Rebuild retrieval with sentence-transformers; re-measure neighbor diagnostics.
4. Run cloud LLM grounded drafts on the 50-pack + golden subsample; compare to copy baseline.
5. Expand IAA beyond 50; adjudicate disagreements into a cleaner gold.
6. Near-duplicate / paraphrase leakage scan beyond exact text.
7. Production sketch: PII redaction, tool permissions, audit log of evidence IDs.

7. Human evaluation work completed

| Work | Annotator | Artifact |

| Golden intent + escalate (N=199) | primary human | `evaluation/golden/golden\_set.csv` |

| Reply quality ratings (N=50) | `annotator\_1` | `evaluation/reply\_human/human\_ratings.csv` |

| Second-pass intent/escalate (N=50) | `annotator\_2` | `evaluation/golden/second\_annotator.csv` |

| LLM-judge agreement | harness + local LLM | `reports/phase2/judge\_human\_agreement.json` |

8. Reproduce headline results (<15 min after data present)

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
python -m src.cli.build_index --embedding tfidf_svd # ~30s
python -m src.cli.eval_phase2 # golden matrix
python -m src.cli.run_agent --mode deterministic \
  --message "Hulu keeps buffering when I try to watch live TV"
make test
```

Optional judge agreement (needs Ollama or OpenAI):

```
python -m src.cli.run_judge --mode ollama
```

Decision log: reports/decision-log.md (condensed list below also in reports/decision-log-short.md).

Decision log (final) — 12 decisions

1. **Decision:** Explicit TWCS acquisition (no silent full download).

Why: Auditability / offline safety.

Alternative: Auto-download on import.

Trade-off: Extra setup step.

Consequence: Reproducible installs without surprise 500MB pulls.

2. **Decision:** Brand = hulu_support by suitability, not max volume.

Why: Cleaner 6–12 intent taxonomy.

Alternative: AmazonHelp.

Trade-off: Less data than mega-brands.

Consequence: Focused evaluation narrative.

3. **Decision:** Conservative conversation IDs for huge multi-author components.

Why: Undirected graphs merged unrelated customers.

Alternative: Always trust components.

Trade-off: Possible over-fragmentation.

Consequence: Safer splits.

4. **Decision:** Conversation-level split + exact duplicate-text purge.

Why: Prevent thread/text leakage.

Alternative: Tweet-level random split.

Trade-off: Slightly smaller eval pools.

Consequence: Honest retrieval/intent metrics.

5. **Decision:** 10 human-reviewed intents from train-only discovery.

Why: Fits Hulu themes without Banking77 sprawl.

Alternative: Raw cluster IDs or 77 banking intents.

Trade-off: Boundary fuzz (playback vs live).

Consequence: Interpretable taxonomy.

6. **Decision:** Keep TF-IDF+LR as primary intent classifier.

Why: Measured baseline; embeddings ≠ automatic upgrade.

Alternative: Embed+linear or LLM classify.

Trade-off: Lexical limits.

Consequence: Semantic agent intent = 0.693/0.668 matches LR.

7. **Decision:** Default embeddings = tfidf_svd.

Why: CI/demo without model downloads.

Alternative: Force sentence-transformers.

Trade-off: Weaker paraphrase retrieval.

Consequence: <15 min headline path.

8. **Decision:** Deterministic escalation policy is final authority.

Why: LLM must not override safety.

Alternative: Model-decided escalate.

Trade-off: Over-escalation.

Consequence: Unsafe auto-handle 0.069 at coverage 0.166.

9. **Decision:** Tune thresholds on validation only.

Why: Golden integrity.

Alternative: Tune on golden.

Trade-off: Noisy validation labels.

Consequence: Honest held-out safety metric.

10. **Decision:** Prefer unsafe auto-handle minimization over coverage.

Why: Trust > automation rate.

Alternative: Maximize auto-handle.

Trade-off: Helpfulness lower on human pack.

Consequence: Headline = unsafe 0.069.

11. **Decision:** LLM judge = evaluator only; not GT.

Why: Groundedness within $\pm 1 = 0.12$ under v1; esc bias.

Alternative: Use judge scores as labels.

Trade-off: Need human pack for reply quality.

Consequence: Reply headlines from annotator_1.

12. **Decision:** Freeze brand/taxonomy/splits/policy for submission.

Why: Stop metric chasing.

Alternative: Keep iterating thresholds on golden.

Trade-off: Known over-escalation remains.

Consequence: Stable submission package.